

A silicon photonic test chip for comparing simulation and measurement

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A $605 \times 410 \mu\text{m}$ test chip combines five Mach–Zehnder interferometers, two single-bus rings, an add-drop ring, an open coupler, and a reference waveguide. Three-dimensional GPU FDTD simulations provide complex scattering matrices from the submitted silicon geometry. SAX connects these matrices into all ten circuits, including through, drop, and backward outputs. Vector mode calculations and SiEPIC EBeam PDK models provide comparisons for dispersion and coupling. The layout supports a common measurement plan: infer group delay from one MZI, test it on the remaining interferometers, then compare ring spectra using the coupling and reference structures. This report gives the simulation results for the forthcoming course measurements.

The chip tests waveguide dispersion and ring coupling within the course floorplan. It uses the SiEPIC EBeam PDK [1], with nominal 500 nm wide, 220 nm thick silicon waveguides, oxide cladding, and quasi-TE excitation near 1550 nm. The reference waveguide and open coupler help interpret the interferometer and ring measurements (Fig. 1, Table 1).

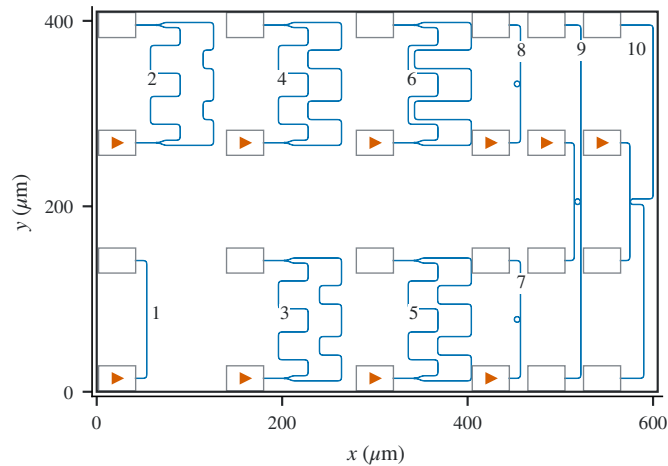


Fig. 1. Submitted mask. Numbers identify the devices in Table 1. Gray boxes are the process-specific grating-coupler black boxes; orange arrows mark laser inputs.

Table 1. Devices and measurement roles.

No.	Structure	Primary comparison
1	REF, $L = 147.897 \mu\text{m}$	Common spectral envelope
2	MZI, $\Delta L = 0$	Splitter/arm baseline
3	MZI, $\Delta L = 50 \mu\text{m}$	FSR holdout
4	MZI, $\Delta L = 100 \mu\text{m}$ A	Group-delay calibration
5	MZI, $\Delta L = 100 \mu\text{m}$ B	Fabrication repeat
6	MZI, $\Delta L = 200 \mu\text{m}$	FSR holdout
7	Single-bus ring, $g = 80 \text{ nm}$	Notch, FSR, linewidth
8	Single-bus ring, $g = 100 \text{ nm}$	Coupling sensitivity
9	Add-drop ring, $g = 80 \text{ nm}$	Through and drop jointly
10	Open half-ring, $g = 80 \text{ nm}$	Nonresonant coupling

Each MZI arm contains ten identical Bézier routing bends with a nominal $5 \mu\text{m}$ corner size. Changing ΔL therefore changes straight length while keeping the bend count fixed. MZI100B repeats MZI100A at another location. All rings have $R = 3 \mu\text{m}$ and zero straight coupling length. A single-bus loop contains one half-ring coupler and a complementary semicircle; the add-drop loop contains two half-ring couplers. Each loop has nominal length $2\pi R$.

The layout contains 74 connected components and 24 grating couplers at $127 \mu\text{m}$ pitch. The four-grating devices use grating 2 as the laser input, counting from zero at the bottom. Device 9 exposes through, drop, and add-monitor signals at gratings 1, 3, and 0. Device 10 exposes through, cross, and backward signals at gratings 1, 0, and 3. The two-grating devices use the lower grating as input. All measurement labels are unique, and local DRC, connectivity, and measurement-coordinate checks report zero violations.

FDTD [2] simulates layer-1 polygons in three dimensions. Each pulse-driven run uses Palik silicon and oxide permittivities at one selected wavelength and exports that frequency alone. Cells are 20 nm in-plane, 10 nm through silicon, and 20–40 nm in exterior oxide. A quasi-TE vertical mirror halves the domain; PMLs are $0.4 \mu\text{m}$. Tidy3D vector modes [3] provide source and detector profiles.

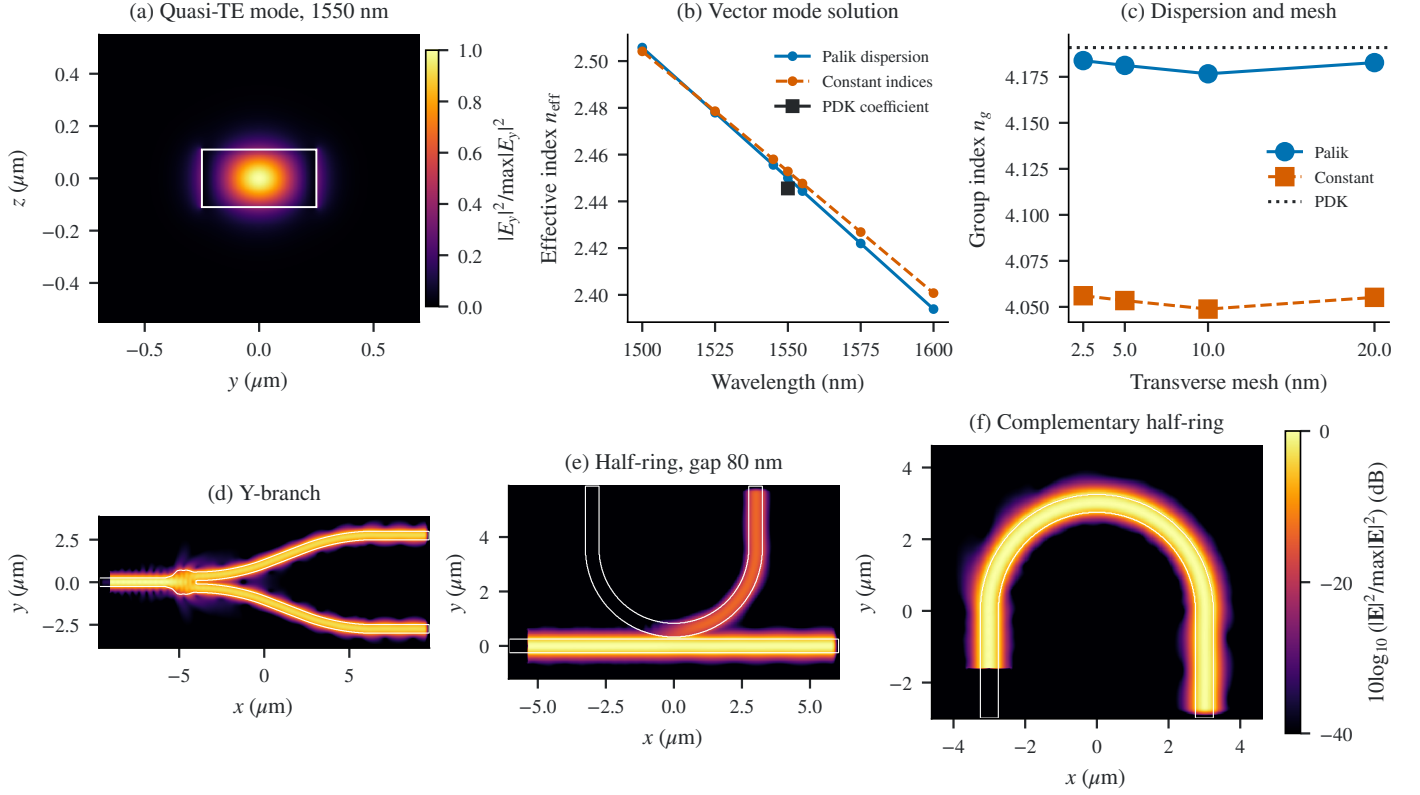


Fig. 2. Mode and component fields. (a) Quasi-TE $|E_y|^2$ profile of the 500×220 nm waveguide at 1550 nm. (b) Effective index versus wavelength for Palik and constant-index materials, with the PDK coefficient. (c) Group index versus transverse mesh. (d) Y-branch splitting. (e) Bus-to-half-ring coupling at an 80 nm gap. (f) Propagation around the complementary $R = 3 \mu\text{m}$ semicircle. Panels (d)–(f) show near-midplane electric-field intensity $|E|^2$ at 1550 nm, with 20 nm in-plane FDTD cells; white contours mark silicon. Each uses its own peak and the same -40 to 0 dB display range. Modal overlaps retain absolute amplitudes.

Thirty calculations cover propagation, all three Y inputs, two inputs of each half-ring coupler, the complementary arc, and the actual R5 corner at 1500, 1550, and 1600 nm. The straight reference adds 1545 and 1555 nm anchors. Access phase is removed at the actual monitor planes, and electric-field parity aligns rotated ports. Dense same-geometry spectra resolve phase branches before interpolation. Independent 1555 nm slices check all components; 1545 nm ring-path excitations check the nearby resonances. Further 10 nm calculations test propagation, both coupling gaps, and the complementary arc. The supplement records the comparisons.

The vector mode study includes Palik material dispersion and width/height variations (Fig. 2). At 1550 nm, its finest Palik mesh (2.5 nm) gives $n_g = 4.1838$; constant indices give 4.0561. The 10 nm FDTD propagation check gives $n_g = 4.1881$, differing by 0.067% from the PDK coefficient 4.19088. Material dispersion affects the MZI fringe spacing,

$$\text{FSR}_{\text{MZI}} \simeq \frac{\lambda^2}{n_g \Delta L}. \quad (1)$$

SAX [4] connects complex component matrices using the layout netlist, following the mode-expansion and S-parameter workflow documented for commercial solvers [5]. With block-diagonal component matrix S_b , internal connection matrix P , and external-port selector E , the independent reference solves

$$S_{\text{ext}} = E^T (I - S_b P)^{-1} S_b E. \quad (2)$$

It agrees with SAX to 4.8×10^{-13} in complex amplitude. The EM routing model cascades straight sections and actual Bézier matrices, including reflections. The ring model retains all coupler ports and counts curved phase once per half-ring. Closed-form MZI and ring checks [6] test the connection algebra separately. Circuit spectra use 20 001 wavelengths, about 5 pm apart.

The Palik GPU model yields raw cross-coupled powers of 0.0439 and 0.0291 at 1550 nm for the 80 and 100 nm gaps, respectively. The common-input Y excess loss is 0.213 dB. Figure 3 and Table 2 compare the resulting circuit spectra.

Table 2. SAX-predicted FSR and half-prominence width (nm). An asterisk marks a line shape that fails the single-Lorentzian fit check.

Observable	PDK S	FDTD S
MZI 50 μm , FSR	11.448	11.448
MZI 100 μm , FSR	5.724	5.734
MZI 200 μm , FSR	2.859	2.859
Ring 80 nm, width	0.250*	0.314
Ring 100 nm, width	0.185*	0.224
Add-drop, through width	0.542	0.553

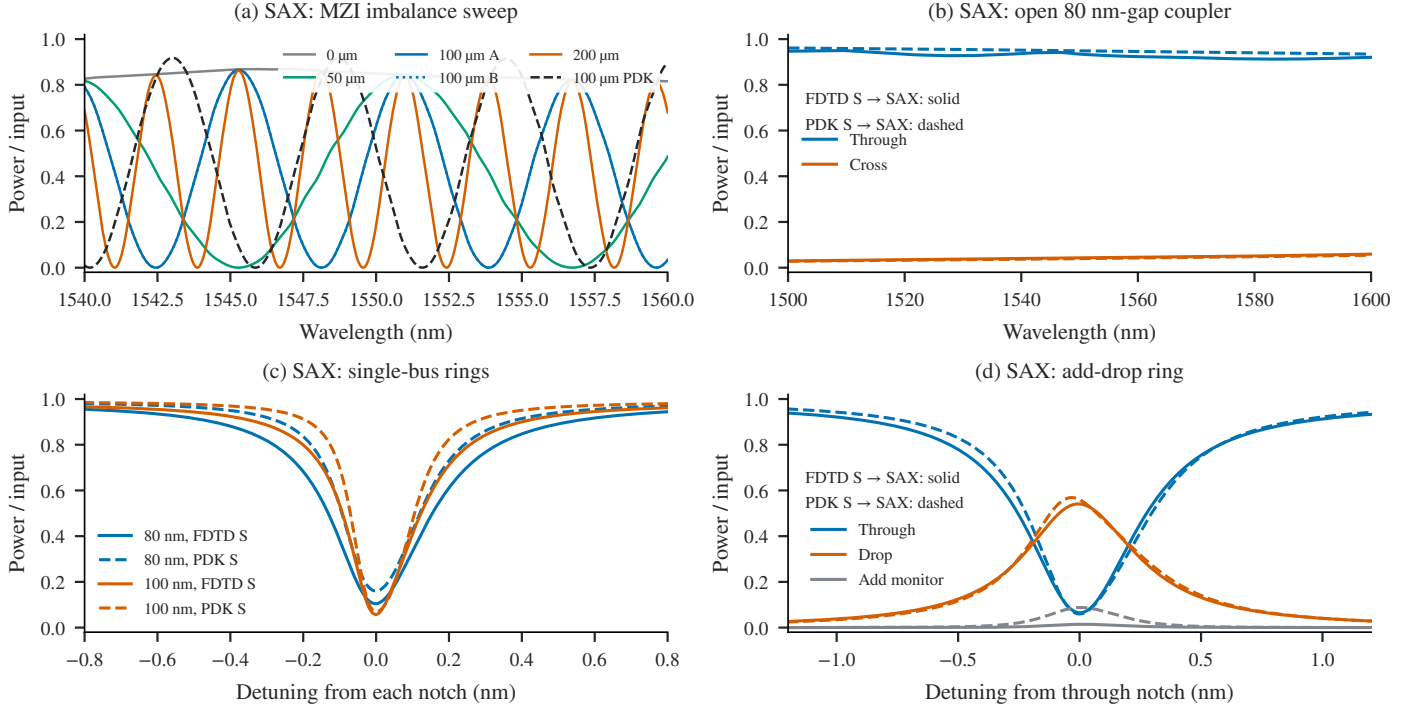


Fig. 3. SAX circuit spectra. (a) MZI transmission for $\Delta L = 0, 50, 100$, and $200 \mu\text{m}$; the $100 \mu\text{m}$ repeat overlaps the original. The black dashed curve uses the PDK model for MZI100A. (b) Open 80 nm-gap coupler through and cross powers. (c) Single-bus ring through spectra for 80 and 100 nm gaps. (d) Add-drop ring through, drop, and add-monitor powers. In (b)–(d), solid curves use 20 nm FDTD S-parameters and dashed curves use PDK S-parameters. Ring detuning is relative to each model's nearby resonance. All powers refer to the on-chip grating tips. Full-band spectra for every device are in the supplement.

The through/drop joint fit gives a shared width of 0.556 nm, with a maximum residual of 1.77% of local contrast. The PDK and 25 nm joint fits exceed the 5% residual threshold. The two output spectra check the fitted line shape.

For MZI100A, the FSR differs by 0.17%, but minima shift by 2.27 nm (Fig. 3). The effective-index offset gives 2.49 rad at 1550 nm over $100 \mu\text{m}$; the 10 nm propagation check reduces the index mismatch.

Plotted models use reciprocal symmetrization and singular-value clipping; raw matrices remain archived. This changes the primary ring-width descriptors by at most 0.60%. Further 10 nm calculations test propagation and both coupling gaps at unchanged slab height. The course data will test absolute resonance wavelengths and fiber losses; process-specific GC geometry arrives during the fabrication merge.

The nominal circuit loss is 3 dB/cm on routed waveguides and the complementary arc. A second PDK scenario uses published historical measurements: 0.3 dB TE Y excess loss at 1550 nm [7] and 7 dB/cm TE spiral loss from September 2014 [8]. These scalar values adjust attenuation while retaining the simulated phase. They provide a measured-data comparison scenario for the design stage.

The measurement analysis will use REF for the common spectral envelope and MZI100A for group delay. MZI50, MZI200, and MZI100B will then test that fitted model. The open coupler constrains splitting after relative output-gain correction. Through and drop spectra will share a fitted resonance center and width, with separate gains and baselines; the add monitor will help identify backward scattering. Coupling and intrinsic loss will be separated where these controls provide enough information.

Only the course dataset is assumed. The PDK default sweep gives a 20.4 pm spacing (5000 points over 1499 to 1601 nm); the analysis will use the actual supplied grid. Resampling checks and the measurement port map are in the supplement.

Data availability. The layout and report are in submissions and reports. Raw simulation data and report sources are retained locally. This report precedes the 2026 fabrication measurements.

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